

DEBUGGING ASSESSEMENT

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1.

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=635`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user's name "Thanishka S" with ID "192521457RA".

On the left sidebar, a task description reads: "11. Create a LL with 5 nodes and display even nodes" with a "1 mark" value and a progress indicator "3/3 answered".

The main content area features a "Debugging Task" instruction: "Find and fix the bugs in the code shown above. Write your corrected code in the editor below." Below this, the "Your Fixed Code" section contains the following C code:

```
1 #include <stdio.h>
2 int linearSearch(int arr[], int n, int key)
3 {
4     int i;
5     for(i = 0; i < n; i++)
6     {
7         if(arr[i] == key)
8             return i;
9     }
10    return -1;
11 }
12 int main()
13 {
14     int n, key, i;
15     scanf("%d", &n);
16     int arr[n];
17     for(i = 0; i < n; i++)
18         scanf("%d", &arr[i]);
19     scanf("%d", &key);
20     printf("%d", linearSearch(arr, n, key));
21     return 0;
22 }
```

To the right of the code editor, the "Test Input" field contains the values: 4, 10 20 30 40, and 20. The "Output" field displays the result: 1. At the bottom of the code editor, there are buttons for "Run", "Save", and "Submit Fix".

The Windows taskbar at the bottom shows the system clock as 2:57 PM on 7/31/2026, along with weather information (37°C, Mostly cloudy) and various application icons.

2.

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar displays "simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=635". The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the sub-header "DS PROGRAMMING", and a user profile for "Thanishka S" with ID "192521457RA". A notification on the left indicates "3/3 answered". The main area is divided into three sections: "Your Fixed Code", "Test Input", and "Output".

Your Fixed Code:

```
1 #include <stdio.h>
2 #define MAX 100
3 int stack[MAX];
4 int top = -1;
5 int pop()
6 {
7     if(top == -1)
8     {
9         printf("Underflow");
10        return -1;
11    }
12    return stack[top--];
13 }
14 int main()
15 {
16     int n, i;
17     scanf("%d", &n);
18     if(n > MAX)
19         return 0;
20     for(i = 0; i < n; i++)
21     {
22         scanf("%d", &stack[i]);
23     }
24     top = n - 1;
25     pop();
26     return 0;
27 }
```

Test Input:

0

Output:

Underflow

3.

The screenshot shows the SIMATS Online Programming Lab interface for a linked list problem. The browser address bar displays "simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=635". The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the sub-header "DS PROGRAMMING", and a user profile for "Thanishka S" with ID "192521457RA". A notification on the left indicates "3/3 answered". The main area is divided into three sections: "Your Fixed Code", "Test Input", and "Output".

Your Fixed Code:

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 struct Node
4 {
5     int data;
6     struct Node *next;
7 };
8 struct Node *createNode(int data)
9 {
10    struct Node *newNode = (struct Node *)malloc(sizeof(struct Node));
11    newNode->data = data;
12    newNode->next = NULL;
13    return newNode;
14 }
15 void display(struct Node *head)
16 {
17     while(head != NULL)
18     {
19         if(head->data % 2 == 0)
20         {
21             printf("%d ", head->data);
22         }
23         head = head->next;
24     }
25 }
26 int main()
27 {
28     struct Node *head = NULL, *temp = NULL;
29     int i, value;
```

Test Input:

10
15
29
26

Output:

Enter 5 node values:
Even nodes: 10 26

SIMATS Online Lab

Not secure simatslab.savitetha.com/practice_app/classactivity/attempt?assignment_id=635

Ask Gemini

Verify it's you

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

Thanishka S 192521457RA

```
27 {
28     struct Node *head = NULL, *temp = NULL;
29     int i, value;
30     printf("Enter 5 node values:\n");
31     for(i = 0; i < 5; i++)
32     {
33         scanf("%d", &value);
34         struct Node *newNode = createNode(value);
35         if(head == NULL)
36         {
37             head = newNode;
38             temp = newNode;
39         }
40         else
41         {
42             temp->next = newNode;
43             temp = newNode;
44         }
45     }
46     printf("Even nodes: ");
47     display(head);
48     return 0;
49 }
```

Run Save Submit Fix

37°C Mostly cloudy

Search

2:58 PM 7/31/2026